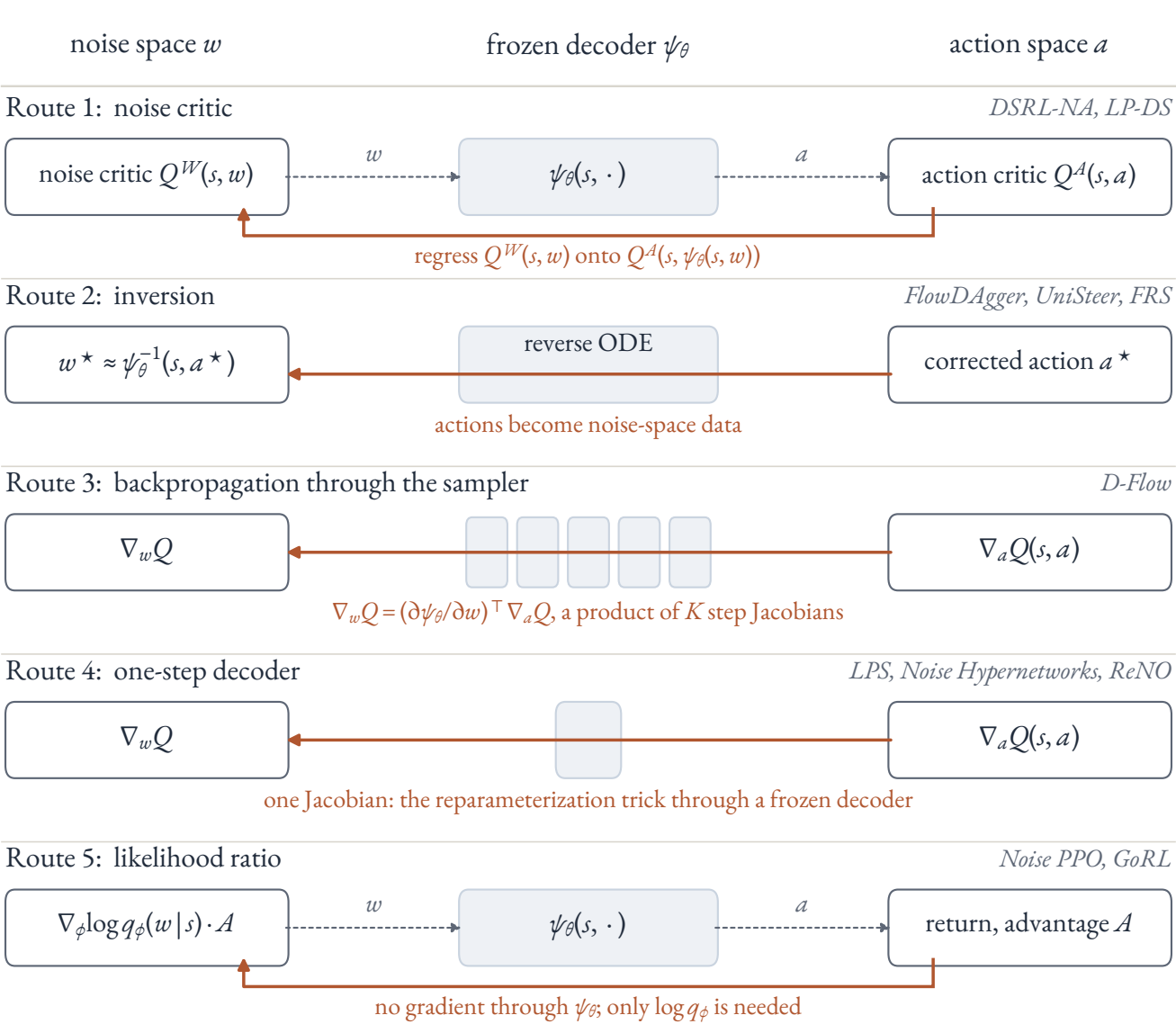
noise space w
frozen decoder ψ_θ
action space a
Route 1: noise critic
DSRL-NA, LP-DS
noise critic $Q^W(s, w)$
w
$\psi_\theta(s, \cdot)$
a
action critic $Q^A(s, a)$
regress $Q^W(s, w)$ onto $Q^A(s, \psi_\theta(s, w))$
Route 2: inversion
FlowDAgger, UniSteer, FRS
$w^\star \approx \psi_\theta^{-1}(s, a^\star)$
reverse ODE
corrected action $a^\star$
actions become noise-space data
Route 3: backpropagation through the sampler
D-Flow
$\nabla_w Q$
$\nabla_a Q(s, a)$
$\nabla_w Q = (\partial \psi_\theta / \partial w)^\top \nabla_a Q$, a product of K step Jacobians
Route 4: one-step decoder
LPS, Noise Hypernetworks, ReNO
$\nabla_w Q$
$\nabla_a Q(s, a)$
one Jacobian: the reparameterization trick through a frozen decoder
Route 5: likelihood ratio
Noise PPO, GoRL
$\nabla_\phi \log q_\phi(w \mid s) \cdot A$
w
$\psi_\theta(s, \cdot)$
a
return, advantage A
no gradient through ψ_θ; only $\log q_\phi$ is needed